

LESSON 2-4 • HOMEWORK PACKET • LEVEL 1 • SHEET 1 OF 2**Building a Box Plot**

Maryland Standard 6.SP.B.4 • Learning target – I can find the five-number summary of a data set and use it to build a box plot.

REMEMBER

- **Five-number summary:** minimum, Q1, median, Q3, maximum.
- **Q1** is the median of the lower half; **Q3** of the upper half. Odd count: leave the median out.
- The **box** goes from Q1 to Q3, with a line at the median. **Whiskers** reach the min and max.

WORKED EXAMPLE

Find the five-number summary of 4, 6, 9, 10, 13, 15, 18.

Step 1 Min = 4, max = 18, median = 10

Step 2 Lower half 4, 6, 9, so Q1 = 6

Step 3 Upper half 13, 15, 18, so Q3 = 15

Answer 4, 6, 10, 15, 18

WORD BANK

five-number summary – min, Q1, median, Q3, max

Q1 – middle of the lower half

Q3 – middle of the upper half

box plot – a box from Q1 to Q3 with whiskers

PRACTICE

show your work • label units

- 1. Push-ups.** Nine students did 22, 40, 10, 34, 18, 46, 25, 14, and 30 push-ups. Find the five-number summary.

Step 1 Put the 9 numbers in order.

Step 2 The median is the 5th number.

Step 3 Find the middle of each half (leave the median out).

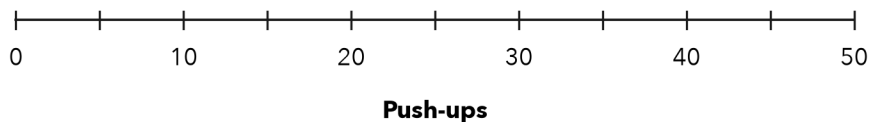
Min	Q1	Median	Q3	Max

- 2.** Draw a box plot of the push-up data from Problem 1.

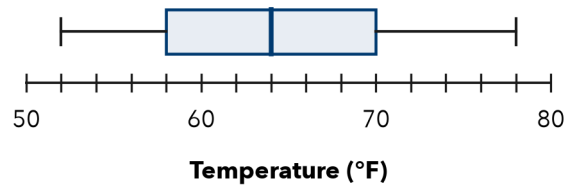
Step 1 Mark the five numbers above the line.

Step 2 Draw the box from Q1 to Q3 and a line at the median.

Step 3 Draw whiskers to the min and max.



- 3. Temperatures.** The box plot shows daily high temperatures ($^{\circ}\text{F}$) for a month.



- a. Median: _____ $^{\circ}\text{F}$
- b. Q1 and Q3: _____
- c. Fraction of days from 58 to 70°F : _____

- 4. Homework Minutes.** 26, 14, 39, 20, 11, 32, 18, 28. Find each value.

Step 1 Order the 8 numbers.

Step 2 Split them into two halves of 4.

Step 3 Find the middle of each half.

- a. Q1: _____ b. Median: _____ c. Q3: _____

- 5. Find the Error.** Grace says Q1 of 2, 5, 7, 8, 12, 14, 20 is 6. She used 2, 5, 7, 8 as the lower half. Explain her error and find Q1.

Frame Grace should leave ____ out. The lower half is ____, so Q1 = ____.

CHECK YOURSELF

- ☐ I can do this on my own ☐ I need a little help ☐ I have a question about #____

AT HOME Explain to someone at home what the box in a box plot shows.

LESSON 2-4 • HOMEWORK PACKET • LEVEL 1 • SHEET 2 OF 2

Comparing Box Plots

Maryland Standard 6.SP.B.4 • Learning target – I can compare two groups by reading the medians and box widths of their box plots.

REMEMBER

- Compare **centers**: use the medians.
- Compare **spread**: box width and range.
- A **narrower box** ($Q3 - Q1$) is more **consistent**.

WORKED EXAMPLE

A: median 18, box 14–22. B: median 24, box 20–26.

Step 1 Center: $24 > 18$, so B scored higher.

Step 2 Box widths: A is $22 - 14 = 8$; B is $26 - 20 = 6$.

Answer B is higher and more consistent

WORD BANK

median – the line inside the box

box width – $Q3 - Q1$

range – $\text{max} - \text{min}$

consistent – close together

PRACTICE

show your work • label units

1. **Reading Time.** Minutes of reading per day for Class A and Class B.

Step 1 Find each median line. Subtract.

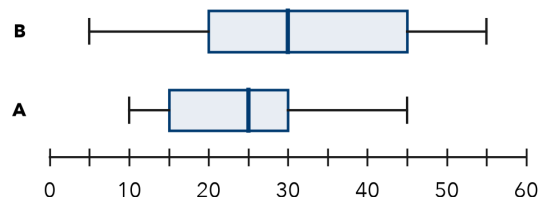
Step 2 Box width = $Q3 - Q1$ for each class.

Step 3 The narrower box is more consistent.

a. Greater median: _____ by _____ min

b. Box widths: A _____ B _____

c. More consistent: _____



2. Find each measure of spread.

a. $Q1 = 12$, $Q3 = 30$. Box width: _____

b. $\text{Min} = 5$, $\text{max} = 41$. Range: _____

3. **Bowling.** Complete the table for two bowlers' scores.

Step 1 Box width = $Q3 - Q1$.

Step 2 Range = $\text{max} - \text{min}$.

	Min	Q1	Median	Q3	Max	Box width	Range
Maya	80	95	102	110	125		
Leon	70	90	105	112	140		

Who is more consistent? _____

4. **Plant Heights.** Both medians are 14 cm. Box S: 12–16 cm. Box T: 8–20 cm. Which varies more?

☐ Group S

☐ Group T

5. **Choose a Bus.** Both medians are 20 min. Boxes: A 18–22, B 14–26 min. Which is more reliable?

Frame Bus ____ is more reliable because its box is ____.

TEST PRACTICE

choose the best answer

6. X: $Q1 = 22$, $Q3 = 34$. Y: $Q1 = 25$, $Q3 = 31$. Which middle half is more consistent?
A X **B** Y **C** They are equal **D** Cannot tell
7. A box plot has the summary 8, 12, 15, 21, 30. How wide is the box?
A 9 **B** 22 **C** 15 **D** 3

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #____